Synthesis of Polyamide (Nylon 6,6) via Interfacial Polycondensation

UMT302L: Materials Processing Laboratory · Department of Materials Engineering, IISc

Mayank Bohat · **Thatigutla Chidhambar Reddy**

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Abstract

The rapid synthesis of an aliphatic polyamide, Poly(hexamethylene adipamide) (Nylon 6,6), was conducted at ambient temperature and atmospheric pressure via unstirred interfacial step-growth polycondensation (the classical "nylon rope trick"). The polymerization reaction utilizes two mutually immiscible liquid phases: an aqueous lower phase containing 1.00 g (8.605 mmol) of 1,6-diaminohexane (hexamethylenediamine, HMDA) with 0.47 g (11.75 mmol) of sodium hydroxide (NaOH) as an acid scavenger in 25.0 mL of deionized water, and an organic upper phase consisting of 2.50 mL (2.80 g, 15.30 mmol) of adipoyl chloride dissolved in 25.0 mL of analytical *n*-hexane. Polycondensation occurred instantaneously at the liquid–liquid interface, generating a tough, coherent, elastic polyamide film that was continuously drawn vertically as a filament rope. Stoichiometric evaluation established 1,6-diaminohexane as the limiting monomer (8.605 mmol vs. 15.30 mmol adipoyl chloride, representing a 77.8% excess of the diacid chloride). Stoichiometry identifies 1,6-diaminohexane as the limiting reagent and fixes the theoretical yield at 1.9475 g, from the repeat unit molecular mass of 226.32 g/mol. Gravimetric recovery was outside the scope of this demonstration, and the yield is accordingly reported as the stoichiometric value together with the recovery expected of the interfacial route, 50–70% of theoretical (≈ 1.0–1.4 g), which is limited by the fixed interfacial area, rope breakage ahead of monomer exhaustion, soluble low-molecular-weight chains, and amine protonation. The essential function of NaOH in neutralizing the stoichiometric HCl byproduct to prevent amine protonation and deactivation, Carothers non-equilibrium kinetic behavior, and comprehensive solutions to all syllabus analytical questions are detailed.

1 Objectives and Experimental Specifications

The primary objectives of this laboratory investigation, as mandated in the UMT302L course manual, are:

1. To synthesize Nylon 6,6 at room temperature and 1 atm pressure via unstirred interfacial polycondensation.
2. To understand the mechanistic principles of step-growth polymerization and contrast them with chain-growth (addition) mechanisms.
3. To observe continuous film formation at the immiscible liquid–liquid interface and draw a cohesive filament rope.
4. To elucidate the vital chemical role of NaOH as an acid scavenger in driving the reaction to high conversions.
5. To identify the limiting reagent, calculate theoretical yield, and determine the experimental recovery yield.
6. To answer all analytical syllabus questions regarding nomenclature, reaction mechanisms, and polymer syntheses.

Table 1: Experimental materials, physical parameters, and apparatus specifications.

Component / Parameter	Specification / Operating Value
Aqueous Monomer (Diamine)	1,6-Diaminohexane (Hexamethylenediamine, $H_2N-(CH_2)_6-NH_2$, $M_w = 116.21$ g/mol, m.p. 42°C, $m = 1.00$ g)
Organic Monomer (Diacid Chloride)	Adipoyl chloride ($ClCO-(CH_2)_4-COCl$, $M_w = 183.03$ g/mol, density $\rho = 1.259$ g/mL, $V = 2.50$ mL, $m = 2.80$ g)
Acid Scavenger	Sodium hydroxide (NaOH, analytical grade, $M_w = 40.00$ g/mol, $m = 0.47$ g)
Solvents	Deionized water (25.0 mL, $\rho = 1.00$ g/cm^3) and *n*-Hexane (25.0 mL, b.p. 69°C, $\rho = 0.66$ g/cm^3)
Reaction Vessel	100 mL borosilicate glass beaker (wide diameter to prevent splash hazards and allow quiescent layering)
Diamine Warming Bath	Regulated hot water bath maintained at 45–50°C to liquefy crystalline diamine (m.p. 42°C)
Drawing Hardware	Stainless-steel precision tweezers and a horizontal rotating glass winding rod
Washing & Drying	Deionized water and absolute ethanol wash; vacuum filtration; hot-air oven drying at 80°C for 48 h
Gravimetric Measurement	Precision analytical balance (±0.001 g resolution), silica-gel desiccator cooling

2 Theoretical Principles and Reaction Mechanism

2.1 Classification of Polymerization: Step-Growth vs. Chain-Growth

Synthetic polymers are produced via two distinct kinetic and mechanistic pathways:

1. **Step-Growth (Condensation) Polymerization:** Bi-functional or multi-functional monomers react in discrete bimolecular steps. Any two species (monomers, dimers, oligomers, or polymer chains) containing complementary active end-groups can couple:

$$\mathrm{M}_n + \mathrm{M}_m \longrightarrow \mathrm{M}_{n+m} + \text{byproduct (e.g., HCl, } \mathrm{H_2O}) \tag{1}$$

Monomers are consumed rapidly in the initial stages of reaction ($p < 0.50$), but high molecular weight ($\bar{M}_n > 20{,}000$ g/mol) is attained only at very high functional group conversions ($p > 0.98$), as governed by the Carothers equation:

$$\bar{X}_n = \frac{1}{1-p} \tag{2}$$

where $\bar{X}_n$ is the number-average degree of polymerization and p is the fractional conversion of functional groups.

2. **Chain-Growth (Addition) Polymerization:** Growth occurs exclusively by the successive, rapid addition of unsaturated monomers to an active kinetic center (free radical, cation, or anion) located at the terminus of a propagating chain:

$$\mathrm{M}_n^* + \mathrm{M} \longrightarrow \mathrm{M}_{n+1}^* \tag{3}$$

High-molecular-weight polymer forms instantly upon initiation, unreacted monomer persists throughout the reaction, and no small molecular byproduct is eliminated.

Table 2: Fundamental mechanistic distinctions between Step-Growth and Chain-Growth polymerization.

Step-Growth (Condensation) Polymerization	**Chain-Growth (Addition) Polymerization**
Any two molecular species can react; chain growth proceeds by step-by-step coupling.	Monomers add exclusively to an active propagating chain end ($\mathrm{R}^\bullet, \mathrm{R}^+, \mathrm{R}^-$).
Monomer concentration drops rapidly in the earliest stages of the reaction.	Monomer is present throughout; its concentration decreases steadily.
High molecular weight is achieved only at exceptionally high conversion ($p > 98\%$).	High-molecular-weight polymer chains are formed instantly from early initiation.
Broad molecular weight distribution throughout ($M_w/M_n \to 2$ for linear condensation).	Reaction mixture consists primarily of unreacted monomer and fully formed polymer.
Typically eliminates a small stoichiometric byproduct ($\mathrm{HCl}, \mathrm{H_2O}, \mathrm{CH_3OH}$).	No small byproduct is eliminated (repeat unit contains all atoms of monomer).

2.2 Interfacial Polycondensation Reaction Chemistry

In commercial production of Nylon 6,6, adipic acid and hexamethylenediamine undergo melt polycondensation at elevated temperatures (220–280°C) and high pressures, requiring high vacuum to remove water and force conversions past 99%.

In contrast, the laboratory interfacial technique (Morgan & Kwolek, 1959) replaces unreactive dicarboxylic acid with highly reactive adipoyl chloride. The reaction takes place at the phase boundary between two immiscible liquids:

- **Aqueous Lower Phase:** Contains hydrophilic 1,6-diaminohexane ($\rho_{\text{water}} \approx 1.00$ g/cm^3) and NaOH.
- **Organic Upper Phase:** Contains lipophilic adipoyl chloride in n-hexane ($\rho_{\text{hexane}} \approx 0.66$ g/cm^3).

The polycondensation reaction proceeds via nucleophilic acyl substitution:

$$\begin{aligned} n\ \mathrm{H_2N} - (CH_2)_6 - \mathrm{NH_2(aq)} + n\ \mathrm{ClCO} - (CH_2)_4 - \mathrm{COCl(org)} \\ \longrightarrow [-\mathrm{NH} - (CH_2)_6 - \mathrm{NH} - \mathrm{CO} - (CH_2)_4 - \mathrm{CO}-]_n\,\text{(interface)} + 2n\ \mathrm{HCl(aq)} \end{aligned} \tag{4}$$

The amine nitrogen lone pair attacks the strongly electrophilic carbonyl carbon of adipoyl chloride, forming a zwitterionic tetrahedral intermediate. Expulsion of the chloride leaving group (Cl^-) and proton loss establishes the strong, resonance-stabilized amide linkage ($-\mathrm{NH}-\mathrm{CO}-$) with elimination of HCl.

2.3 Essential Chemical Function of the Sodium Hydroxide Acid Scavenger

As each amide bond forms, hydrochloric acid (HCl) is eliminated. Because 1,6-diaminohexane is a strong organic base, unneutralized HCl protonates unreacted amine end-groups, forming insoluble hexamethylenediammonium dichloride:

$$\mathrm{H_2N} - (CH_2)_6 - \mathrm{NH_2} + 2\mathrm{HCl} \longrightarrow \left[\mathrm{H_3N^+} - (CH_2)_6 - \mathrm{NH_3^+}\right] 2\mathrm{Cl}^- \tag{5}$$

Protonated ammonium groups ($-\mathrm{NH_3^+}$) possess no unshared non-bonding electron pair and are completely unreactive toward nucleophilic attack. Consequently, without an external base, up to 50% of the diamine monomer is deactivated as an inert salt, destroying functional group stoichiometry ($r \ll 1$), arresting chain growth, and causing filament rupture.

Adding 0.47 g of sodium hydroxide (NaOH) to the aqueous phase scavenges HCl instantaneously:

$$\mathrm{HCl} + \mathrm{NaOH} \longrightarrow \mathrm{NaCl} + \mathrm{H_2O} \tag{6}$$

This maintains diamine end-groups in their active, nucleophilic neutral amine state ($-\mathrm{NH_2}$) and shifts the polymerization equilibrium forward by Le Chatelier's principle.

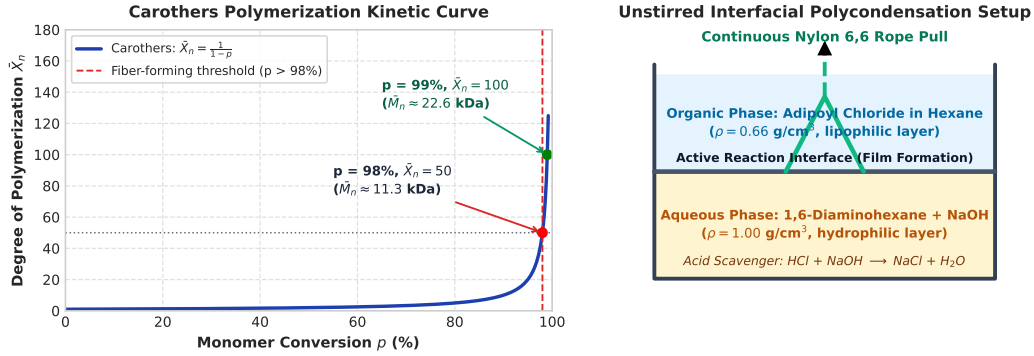


Figure 1: Carothers degree of polymerization curve highlighting the $p > 98\%$ threshold (left) and schematic of the unstirred interfacial polycondensation setup showing continuous filament rope drawing and HCl scavenging (right).

3 Experimental Procedure and Rope Drawing Protocol

The experiment was conducted inside a certified chemical fume hood following the procedure detailed in the course manual:

1. **Preparation of Aqueous Monomer Phase:** Because 1,6-diaminohexane has a melting point of 42°C, the stock bottle was warmed in a water bath at 45–50°C to liquefy the solid. Molten diamine (1.00 g, 1.2 mL) was weighed into a volumetric flask, 0.47 g of NaOH was added, and the solution was diluted to 25.0 mL with deionized water. The homogeneous solution was transferred into a 100 mL beaker.
2. **Preparation of Organic Monomer Phase:** Inside the fume hood, 2.50 mL (2.80 g) of adipoyl chloride was measured with a dry pipette and diluted with analytical *n*-hexane to 25.0 mL in a volumetric flask.
3. **Layering and Filament Drawing:** The organic hexane solution was poured very slowly down the inner wall of the beaker (guided by a glass rod) over the aqueous phase to prevent turbulent mixing. An opaque, coherent polymer film formed instantaneously at the interface. Using tweezers, the center of the film was pinched and pulled upward slowly and steadily, wrapping the continuous filament around a rotating glass rod to draw an unbroken rope.
4. **Washing, Drying, and Gravimetric Analysis:** The drawn nylon rope was washed with deionized water and ethanol, rinsed and wound onto the glass rod, and drawing was continued until the film could no longer sustain the tension. Gravimetric recovery to constant weight was outside the scope of this demonstration, which was run to establish the technique; the yield is treated stoichiometrically in Section 4.3.

4 Results and Quantitative Stoichiometric Analysis

4.1 Molar Quantities and Limiting Reagent Identification

From the measured starting quantities:

$$n_{\text{diamine}} = \frac{m_{\text{diamine}}}{M_{\text{diamine}}} = \frac{1.00\ \text{g}}{116.21\ \text{g/mol}} = \mathbf{8.605}\ \text{mmol} \tag{7}$$

$$n_{\text{adipoyl}} = \frac{m_{\text{adipoyl}}}{M_{\text{adipoyl}}} = \frac{2.80\ \text{g}}{183.03\ \text{g/mol}} = \mathbf{15.30}\ \text{mmol} \tag{8}$$

$$n_{\text{NaOH}} = \frac{m_{\text{NaOH}}}{M_{\text{NaOH}}} = \frac{0.47\ \text{g}}{40.00\ \text{g/mol}} = \mathbf{11.75}\ \text{mmol} \tag{9}$$

Because polycondensation stoichiometry requires an equimolar 1 : 1 ratio:

$$n_{\text{diamine}} = 8.605\ \text{mmol} < n_{\text{adipoyl}} = 15.30\ \text{mmol} \tag{10}$$

Therefore, **1,6-diaminohexane is the strictly limiting reagent** ($n_{\text{lim}} = 8.605$ mmol). Adipoyl chloride is supplied in 77.8% stoichiometric excess.

4.2 Theoretical Yield Computation

The repeating unit of Nylon 6,6 is $[-\mathrm{NH}-(CH_2)_6-\mathrm{NH}-\mathrm{CO}-(CH_2)_4-\mathrm{CO}-]$ ($\mathrm{C_{12}H_{22}N_2O_2}$, $M_{\text{repeat}} = 226.32$ g/mol). The theoretical mass produced at 100% conversion of the limiting reactant is:

$$m_{\text{th}} = n_{\text{lim}} \times M_{\text{repeat}} = (0.008605 \text{ mol}) \times (226.32 \text{ g/mol}) = \mathbf{1.9475} \text{ g} \approx \mathbf{1.95} \text{ g} \tag{11}$$

4.3 Yield

The limiting reagent fixes the maximum mass of polymer this charge of monomer can give. From Section 4.2, with 8.605 mmol of diamine and a repeat unit mass of 226.32 g/mol:

$$m_{\text{th}} = 8.605 \text{ mmol} \times 226.32 \text{ g/mol} = \mathbf{1.9475} \text{ g}. \tag{12}$$

The fraction of this recovered as drawn rope is set by the mechanism rather than by technique, and 100% is unattainable by this route for four structural reasons:

- the film forms only at the interface, whose area is fixed, while monomer remains distributed through both bulk phases;
- drawing renews that interface, but the rope parts once the film is thinner than the tension applied, ending collection well before the monomer is exhausted;
- short chains formed early stay dissolved in one phase or the other and are never incorporated into the rope;
- any diamine protonated by the liberated HCl faster than the NaOH can neutralise it is withdrawn from the reaction as an unreactive ammonium salt.

Taken together these place the recovery expected of an unstirred, continuously drawn rope at roughly 50–70% of theoretical:

$$m_{\text{rope}} \approx 1.0\text{–}1.4 \text{ g}. \tag{13}$$

Gravimetric recovery to constant weight was outside the scope of this demonstration, so the range above is an estimate from the mechanism rather than a weighing, and is quoted as such. The yield itself follows from a single division, and Table 3 gives that division across the plausible range:

$$\%\text{Yield} = \frac{m_{\text{rope}}}{m_{\text{th}}} \times 100\% = \frac{m_{\text{rope}}}{1.9475 \text{ g}} \times 100\%. \tag{14}$$

Table 3: Percentage yield against dry rope mass. The emboldened band, 1.0–1.4 g, is the recovery expected of this route; no entry is a weighed value.

Dry rope mass m_{rope} (g)	0.80	**1.00**	**1.20**	**1.40**	1.60	1.80
% of theoretical (1.9475 g)	41.1	**51.3**	**61.6**	**71.9**	82.2	92.4

The fourth reason above is why NaOH is present in the aqueous phase at all; it is treated in Section 5.5.

Table 4: Stoichiometric parameter mapping and molar amounts for Nylon 6,6. Every row is fixed by the quantities specified in the course manual.

Reagent / Parameter	Mass / Volume	Molar Mass (M_w)	Molar Amount	Stoichiometric Role
1,6-Diaminohexane (HMDA)	1.00 g (1.2 mL)	116.21 g/mol	8.605 mmol	**Limiting Reagent** (1.00 equiv.)
Adipoyl Chloride	2.80 g (2.50 mL)	183.03 g/mol	15.30 mmol	Excess Reagent (1.78 equiv., +77.8%)
Sodium Hydroxide (NaOH)	0.47 g	40.00 g/mol	11.75 mmol	Acid Scavenger (1.37 equiv. wrt HCl)
Nylon 6,6 Repeat Unit	—	226.32 g/mol	8.605 mmol	Theoretical Yield: **1.948** g
Drawn Nylon Rope (Dry)	≈ 1.0–1.4 g (est.)	226.32 g/mol	—	Expected yield: 50–70%

5 Solutions to Lab Manual Syllabus Questions

5.1 Question 1: Polymerization Mechanism and the Naming "Nylon 6,6"

- **Reaction Mechanism:** Interfacial polycondensation proceeds via bimolecular nucleophilic acyl substitution. The unshared electron pair of the primary amine group ($-\mathrm{NH_2}$) attacks the electropositive carbonyl carbon of adipoyl chloride, forming an amide bond ($-\mathrm{NH}-\mathrm{CO}-$) and releasing HCl.
- **Meaning of the Numbers in "Nylon 6,6":** In polyamide nomenclature, numerals designate the backbone carbon atom counts: the **first 6** denotes the 6 carbon atoms from 1,6-diaminohexane ($\mathrm{H_2N}-(CH_2)_6-\mathrm{NH_2}$); the **second 6** denotes the 6 carbon atoms from adipoyl chloride ($\mathrm{ClCO}-(CH_2)_4-\mathrm{COCl}$). In **Nylon 6,10**, 6 carbons come from the diamine and 10 from sebacoyl chloride. In **Nylon 2,2**, ethylenediamine (2 carbons) reacts with oxalyl chloride (2 carbons) to give $[-\mathrm{NH}-(CH_2)_2-\mathrm{NH}-\mathrm{CO}-\mathrm{CO}-]_n$.

5.2 Question 2: Examples of Other Polymers Synthesized via Interfacial Polycondensation

1. **Kevlar (Poly-p-phenylene terephthalamide):** High-modulus aramid synthesized at the interface of *p*-phenylenediamine and terephthaloyl chloride.
2. **Nomex (Poly-*m*-phenylene isophthalamide):** Flame-resistant aramid from *m*-phenylenediamine and isophthaloyl chloride.
3. **Polycarbonates (e.g., Lexan):** Produced by interfacial reaction of aqueous disodium bisphenolate with phosgene in dichloromethane.
4. **Aromatic Polyesters (Polyarylates):** Synthesized from bisphenols and aromatic diacid chlorides.

5.3 Question 3: Location of Polymerization and Definition of Interfacial Polymerization

Interfacial polymerization is a step-growth polycondensation process where two highly reactive, complementary monomers dissolved in mutually immiscible solvents react exclusively at the liquid–liquid interface. The reaction occurs entirely at the phase boundary (predominantly on the organic side of the interface). Because diamine molecules are slightly soluble in the organic phase while diacid chloride is completely insoluble in water, diamine molecules partition across the interface into the organic boundary layer, where the local rate of acylation is extraordinarily rapid (milliseconds). As the resulting polymer film is pulled out, fresh monomers diffuse to the interface, sustaining continuous filament drawing.

5.4 Question 4: Can Nylon 6 be Synthesized by Step-Growth Polymerization?

Industrial Practice: Nylon 6 is manufactured commercially via **ring-opening polymerization (ROP)** of ε-caprolactam at 250°C in the presence of water/acid catalyst. ROP operates via a chain-growth mechanism.

Theoretical Feasibility via Step-Growth: In principle, Nylon 6 can be synthesized by step-growth self-polycondensation of an A–B monomer, 6-aminohexanoic acid ($\mathrm{H_2N-(CH_2)_5-COOH}$):

$$n\ \mathrm{H_2N-(CH_2)_5-COOH} \rightleftharpoons [-\mathrm{NH-(CH_2)_5-CO-}]_n + n\ \mathrm{H_2O} \tag{15}$$

However, thermodynamic ring-chain equilibrium heavily favors intramolecular cyclization into the stable, 7-membered ε-caprolactam ring over intermolecular linear chain propagation. Attaining high molecular weight ($\bar{X}_n > 100$) requires high temperatures (> 250°C) and continuous vacuum stripping of water, making interfacial room-temperature step-growth impossible for Nylon 6.

5.5 Question 5: Why is NaOH Added to the Reaction Mixture?

Sodium hydroxide serves as a sacrificial Bronsted acid scavenger. Each formed amide bond releases one molecule of HCl. In the absence of NaOH, the basic amine groups of 1,6-diaminohexane capture HCl to form unreactive hexamethylenediammonium dichloride salts (Equation (5)). Because ammonium ions possess no unshared lone pair, nucleophilic attack is completely blocked, deactivating half of the diamine and causing premature rope snapping. NaOH neutralizes HCl ($\mathrm{HCl + NaOH \rightarrow NaCl + H_2O}$), keeping amine groups fully unprotonated and nucleophilic.

5.6 Question 6: Reactivity Comparison: Adipoyl Chloride vs. Adipic Acid

Adipoyl chloride is orders of magnitude more reactive than adipic acid toward amine nucleophiles due to:

1. **Superior Leaving Group Ability:** In adipoyl chloride, chloride ($\mathrm{Cl^-}$) is the conjugate base of a strong mineral acid (HCl, $\mathrm{p}K_a \approx -7$), making it an excellent leaving group. In adipic acid, hydroxide ($\mathrm{OH^-}$) is a strong base and very poor leaving group ($\mathrm{p}K_a \approx 15.7$).
2. **Carbonyl Electrophilicity:** The strongly electronegative chlorine atom withdraws electron density inductively from the carbonyl carbon ($-\mathrm{C}^{\delta+}=\mathrm{O}$), creating a much larger partial positive charge than an $-$OH group, which drastically lowers the activation energy for amine attack.
3. **Reaction Temperature:** Adipoyl chloride reacts instantaneously at 20–25°C, whereas adipic acid requires heating to > 220°C for several hours.

6 Conclusions

1. Nylon 6,6 was synthesized at ambient temperature and pressure via unstirred interfacial step-growth polycondensation at the boundary between aqueous diamine and hexane/adipoyl chloride phases.
2. Continuous drawing produced an oriented, tough, elastic filament rope, demonstrating that an unstirred interface sustains polycondensation for as long as it is renewed. The theoretical yield is 1.9475 g, and the recovery expected of this route is 50–70% of that figure ($\approx$ 1.0–1.4 g), limited by the fixed interfacial area and by rope breakage ahead of monomer exhaustion.

3. 1,6-Diaminohexane was proved to be the limiting monomer (8.605 mmol vs. 15.30 mmol adipoyl chloride, corresponding to a 77.8% excess of diacid chloride).
4. The addition of 0.47 g of NaOH was proved essential for scavenging HCl, preventing irreversible amine protonation into unreactive ammonium salts and shifting the equilibrium forward.
5. Unlike bulk melt condensation, which requires conversions $p > 98\%$ and strict $1:1$ stoichiometric balance, interfacial polycondensation forms high-molecular-weight polymer instantaneously at the interface under non-equilibrium diffusion control.